Chapter 25: Geometric Optics

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PHY201

Lecture 25

Part I

The Ray Model of Light

Outline of Part I: The Ray Model of Light

Outline of Part I: The Ray Model of Light

What Is Geometric Optics?

Geometric optics models light as **rays** — straight-line paths that carry energy through space.

The Central Question

- How do rays propagate through optical systems?
- How do they change direction at boundaries?
- How do they form images?

Fermat's Principle

Light travels along the path of *least time.*

Reflection, refraction, and image formation all emerge naturally from this single

When Does Geometric Optics Work?

The ray approximation requires:

$$\text{object size} \gg \lambda$$

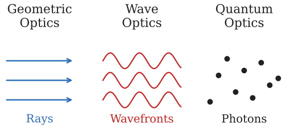
All optical elements must be much larger than the wavelength of light.

Outline of Part I: The Ray Model of Light

Introduction to Geometric Optics

Three Models of Light

Three Complementary Models of Light



Three Regimes

- **Geometric optics** — rays, image formation
- **Wave optics** — interference, diffraction
- **Quantum optics** — photons, transitions

Not Competing Theories

Each model is a valid *approximation* at a different physical scale.

Camera lens → geometric.
Soap bubble → wave.

When the Ray Model Succeeds — and When It Fails

Model Applicability

Situation	Ray Model?	Why
Camera lens	Excellent	Lens $\gg \lambda$
Shadow behind a doorway	Approximate	Edge diffraction occurs
Soap-bubble colors	Poor	Interference dominates

The Ray Approximation

In a **uniform** medium, light travels in straight lines.

Direction changes *only* at boundaries between media.

Visible light: $\lambda \approx 400$–$700\,\text{nm}$.
A magnifying glass: $\sim 5\,\text{cm}$.

Models are tools, not universal truths.

Choosing the right model for the right scale is itself a physics skill.

Check Your Understanding

A magnifying glass forms an image of a small object. Which model of light best describes this process?

Come to lecture to see the answer.

Part II

Reflection

Outline of Part II: Reflection

Outline of Part II: Reflection

The Law of Reflection

Specular and Diffuse Reflection

Light at a Boundary

Whenever light reaches a **boundary** between two media, it simultaneously *reflects* (stays in the original medium) and *refracts* (enters the new medium).

Organizing Principle

- **Reflection** governs mirrors
- **Refraction** governs lenses and prisms
- **Total internal reflection** is the special case where refraction fails

The normal is a line drawn *perpendicular* to the surface at the point of incidence.

If a ray strikes at $35°$ from the surface, its angle from the normal is $90° - 35° = 55°$.

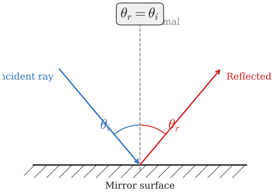
Important convention: all angles are measured from the *surface normal*, not from the surface itself.

The Law of Reflection

Law of Reflection — (cp2e_ch25_s2)

$$\boxed{\theta_r = \theta_i}$$

Both angles measured from the **normal** to the surface.



Variable Definitions

- θ_i: incidence angle (from normal)
- θ_r: reflection angle (from normal)
- All three lines lie in the same plane
- **Error:** angles from *normal*, not

Outline of Part II: Reflection

Specular vs. Diffuse Reflection

Specular Reflection

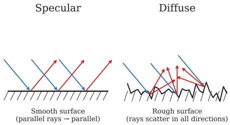
- Smooth surface
- Parallel rays remain parallel after reflection
- Clear image forms

Examples: mirror, calm water, polished metal

Diffuse Reflection

- Rough surface
- Surface normals point in different directions
- Reflected rays scatter in all directions
- No image forms

Examples: paper, painted wall, wood



Each tiny patch of a rough surface *still obeys* the law of reflection locally.

Diffuse reflection is not a violation — it results from many different local normals.

Image Formation in a Plane Mirror

A plane mirror forms a **virtual image**: rays appear to diverge from a point *behind* the mirror, but no light actually converges there.

Plane Mirror Image Properties

- **Virtual** (behind the mirror)
- **Upright**
- **Same size** as the object
- Located the **same distance behind** the mirror as the object is in front

Real vs. Virtual Images

Real image: light rays *actually converge* at that location — can be projected on a screen.

Virtual image: rays only *appear* to come from that point when traced backward — cannot be projected.

Check Your Understanding

A ray strikes a mirror at $35°$ measured from the *surface*. What is the reflected angle relative to the normal?

Come to lecture to see the answer.

Check Your Understanding

Why can you see your reflection in a calm lake but not in choppy water?

Come to lecture to see the answer.

Part III

Refraction and the Index of Refraction

Outline of Part III: Refraction and the Index of Refraction

Outline of Part III: Refraction and the Index of Refraction

The Index of Refraction

Index of Refraction n — (cp2e_ch25_s3)

$$\boxed{n = \frac{c}{v}}$$

$c = 3.00 \times 10^8\ \frac{\mathrm{m}}{\mathrm{s}}$ (vacuum speed); $v =$ speed in medium ($v \leq c$).

Typical Values

Vacuum	$n = 1.000$
Water	$n = 1.333$
Crown glass	$n = 1.52$
Diamond	$n = 2.419$

Physical Meaning

$n = 1$: vacuum (fastest)

$n > 1$: all real materials

Larger $n \Rightarrow$ slower v

"Optically denser" = larger n.

n is dimensionless and always ≥ 1 for

Outline of Part III: Refraction and the Index of Refraction

Index of Refraction

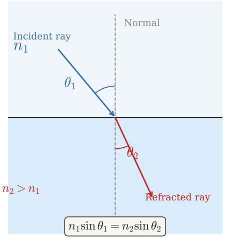
Snell's Law

Snell's Law of Refraction

Snell's Law — (cp2e_ch25_s3)

At a boundary between media 1 and 2:

$$\boxed{n_1 \sin\theta_1 = n_2 \sin\theta_2}$$



Key Variables

- n_1, n_2: refractive indices
- θ_1: incidence angle (from normal)
- θ_2: refraction angle (from normal)
- $n_2 > n_1$: bends *toward* normal
- $n_2 < n_1$: bends *away* from normal

Why Does Light Bend at a Boundary?

When a wavefront crosses a boundary at an angle, one side enters the slower medium *first* — that side slows while the other continues at the original speed, tilting the wavefront.

Wavefront Picture

- Rays are *perpendicular* to wavefronts
- Unequal speeds across the wavefront tilt it
- The tilt of the wavefront *is* the refraction
- This is why Snell's law is geometrically exact

Analogy

Think of a car that crosses from pavement onto mud at an angle: the wheels that hit the mud first slow down, turning the car.

Light bends for exactly the same geometric reason.

Example: Speed of Light in Zircon — (cp2e_ch25_ex1)

Speed of Light in Zircon

Zircon has an index of refraction $n = 1.923$. Find the speed of light in zircon.

$$n = \frac{c}{v}$$

$$v = \frac{c}{n}$$

$$v = \frac{3.00 \times 10^8 \ \frac{\text{m}}{\text{s}}}{1.923}$$

$$\boxed{v = 1.56 \times 10^8 \ \frac{\text{m}}{\text{s}}}$$

Light travels at roughly *half* its vacuum speed inside zircon.

- Is $v < c$? ✓
- Is $n > 1$? ✓
- Higher $n \Rightarrow$ slower speed. Consistent.

Example: Identifying a Liquid by Refraction — (cp2e_ch25_ex2)

Identifying a Liquid

Light passes from air ($n_1 = 1.00$) into an unknown liquid. The incident angle is $\theta_1 = 30.0°$ and the refracted angle is $\theta_2 = 22.0°$. Find the index of refraction of the liquid.

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

$$n_2 = \frac{n_1 \sin\theta_1}{\sin\theta_2}$$

$$n_2 = \frac{(1.00)\sin 30.0°}{\sin 22.0°}$$

$$\boxed{n_2 \approx 1.33}$$

$n \approx 1.33$ matches **water** in the index table.

The ray bent toward the normal ($22° < 30°$) since $n_2 > n_1$ — consistent.

- Did the ray bend toward or away from the normal?
- Does $n_2 > n_1$ predict this? ✓

Check Your Understanding

When light travels from air into glass, does it speed up or slow down? Does the ray bend toward or away from the normal?

Come to lecture to see the answer.

Part IV

Total Internal Reflection

Outline of Part IV: Total Internal Reflection

Outline of Part IV: Total Internal Reflection

The Critical Angle

Applications of TIR

Total Internal Reflection (TIR)

Two Conditions for TIR

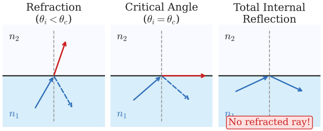
- $n_1 > n_2$ (denser $\rightarrow$ less-dense medium)
- $\theta_i > \theta_c$ (exceeds critical angle)

Why TIR Occurs

At the critical angle, $\theta_2 = 90°$ (refracted ray grazes boundary).

Beyond θ_c: $\sin\theta_2 > 1$ is impossible $\Rightarrow$ all energy reflects.

Going from denser to less-dense medium is required. Light going the other way cannot undergo TIR.



The Critical Angle

Critical Angle θ_c — (cp2e_ch25_s4)

Derived from Snell's law with $\theta_2 = 90°$:

$$\boxed{\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right)}$$

Requires $n_1 > n_2$ (otherwise TIR is impossible).

Variable Definitions

- θ_c: critical angle
- n_1: index of the incident medium (larger)
- n_2: index of the transmitted medium (smaller)

Reference Values

- Water–air: $\theta_c = 48.6°$
- Glass–air: $\theta_c \approx 41°$
- Diamond–air: $\theta_c \approx 24.4°$

Example: Critical Angle for Water — (cp2e_ch25_ex3)

Critical Angle: Water to Air

Find the critical angle for light traveling from water ($n_1 = 1.333$) into air ($n_2 = 1.000$).

$$\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right)$$

$$\theta_c = \sin^{-1}\left(\frac{1.000}{1.333}\right)$$

$$\boxed{\theta_c = 48.6^\circ}$$

Any ray hitting the water–air surface at more than 48.6° from the normal undergoes TIR.

An observer underwater sees the *entire* above-water world compressed into a cone of $2 \times 48.6^\circ \approx 97^\circ$.

- Could TIR occur going from air into water?

Outline of Part IV: Total Internal Reflection

Applications of TIR

Optical Fibers

- **Core:** high-index glass ($n \approx 1.5$)
- **Cladding:** slightly lower-index glass
- Rays hitting core–cladding boundary at $\theta > \theta_c$ undergo TIR
- *Uses:* internet, endoscopy, sensing

Diamond Brilliance

Diamond: $n = 2.419$, $\theta_c \approx 24.4°$.

Tiny θ_c means rays bounce internally many times before exiting — producing brilliance.

A single optical fiber can carry millions of simultaneous conversations via different wavelengths of light.

Diamond cutters design facet angles specifically to maximize TIR toward the viewer's eye.

Check Your Understanding

Which of the following transitions can produce total internal reflection?

(a) Air $\rightarrow$ glass (b) Water $\rightarrow$ diamond (c) Glass $\rightarrow$ air (d) Vacuum $\rightarrow$ water

Come to lecture to see the answer.

Part V

Dispersion, Prisms, and Rainbows

Outline of Part V: Dispersion, Prisms, and Rainbows

Dispersion

Outline of Part V: Dispersion, Prisms, and Rainbows

Dispersion

Dispersion: Color and the Index of Refraction

Dispersion is the variation of the index of refraction with wavelength. Different colors have slightly different values of n, so they refract by different amounts.

Color Ordering in Glass

- **Violet**: shorter λ, *larger* n, bends more
- **Red**: longer λ, *smaller* n, bends less
- White light separates into a continuous spectrum

In **vacuum**: all colors travel at c — no dispersion.

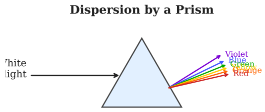
In **matter**: different colors travel at different speeds — dispersion occurs.

Crucial Clarification

Frequency stays constant at a boundary.

Wavelength changes because speed changes: $\lambda = v/f$.

Prisms and Rainbow Formation



Rainbows

In each droplet: refraction in → internal reflection → refraction out → color separation.

Angles (from anti-solar point):

- Red: ≈ 42° (outer arc)
- Violet: ≈ 40° (inner arc)

Check Your Understanding

Which color of visible light experiences the greatest deflection when passing through a glass prism?

Come to lecture to see the answer.

Part VI

Image Formation by Thin Lenses

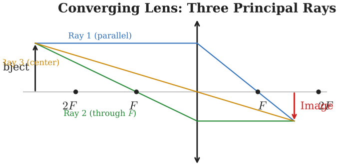
Outline of Part VI: Image Formation by Thin Lenses

Outline of Part VI: Image Formation by Thin Lenses

Converging and Diverging Lenses

Converging Lens ($f > 0$)

- Thicker at the center
- Parallel rays converge to a real focal point
- Also called a *convex* lens
- Applications: camera, eye, projector



Diverging Lens ($f < 0$)

- Thinner at the center
- Parallel rays diverge as if from a virtual focal point
- Also called a *concave* lens
- Applications: eyeglasses for myopia, viewfinders

Both lens types form images through refraction at curved surfaces.

The *sign* of f distinguishes converging from diverging.

Outline of Part VI: Image Formation by Thin Lenses

Principal Ray Tracing for a Converging Lens

Three Principal Rays (use any two)

1. Ray **parallel** to the optical axis $\rightarrow$ refracts through the **far focal point** F
2. Ray through the **near focal point** F $\rightarrow$ emerges **parallel** to the axis
3. Ray through the **optical center** $\rightarrow$ passes through **undeviated**

Image Location

The image forms where any two principal rays intersect (real image) or appear to diverge from (virtual image).

Real vs. Virtual

Real image: rays actually converge — $d_i > 0$.

Virtual image: rays appear to diverge

Key Insight

A converging lens forms a virtual image only when the object is *inside* the focal length.

A diverging lens *always* forms a virtual image for a real object.

Outline of Part VI: Image Formation by Thin Lenses

The Thin Lens Equation

Thin Lens Equation and Magnification — (cp2e_ch25_s6)

$$\boxed{\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}} \qquad m = \frac{h_i}{h_o} = -\frac{d_i}{d_o}$$

Variable Definitions

- d_o: object distance (> 0 for real objects)
- d_i: image distance
- f: focal length
- m: lateral magnification

Sign Conventions

- $d_i > 0$: real image (far side)
- $d_i < 0$: virtual image (same side)
- $f > 0$: converging lens
- $f < 0$: diverging lens
- $m > 0$: upright; $m < 0$: inverted

Lens Power and Image Summary

Converging Lens

Object	**Image**
Beyond $2f$	Real, inv., reduced
At $2f$	Real, inv., same size
Btwn f–$2f$	Real, inv., enlarged
Inside f	Virtual, upright, enl.

Lens Power P

$$\boxed{P = \frac{1}{f}}$$

f in m; P in diopters (D). *Eyeglass Rx in* D.

Diverging Lens: Always

- Virtual
- Upright
- Reduced
- Same side as object

Virtual image cannot be projected.

Real image ($d_i > 0$) can be projected.

Example: Converging Lens Image — (cp2e_ch25_ex4)

Image Location and Magnification

A converging lens has focal length $f = 20.0\,\text{cm}$. An object is placed $d_o = 30.0\,\text{cm}$ from the lens. Find the image distance and magnification.

$$\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o}$$

$$\frac{1}{d_i} = \frac{1}{20.0} - \frac{1}{30.0}$$

$$\frac{1}{d_i} = \frac{3-2}{60.0} = \frac{1}{60.0}$$

$$\boxed{d_i = 60.0\,\text{cm}}$$

$$\boxed{m = -\frac{d_i}{d_o} = -\frac{60.0}{30.0} = -2.00}$$

$d_i > 0$: **real** image on far side.

$m < 0$: **inverted**.

$|m| = 2$: **enlarged** to twice the object size.

- Object is between f and $2f$ (20 cm–40 cm). Consistent with real, inverted, enlarged? ✓

Example: Magnifying Glass — (cp2e_ch25_ex5)

Object Inside the Focal Length

A converging lens has $f = 15.0\,\text{cm}$. An object is placed $d_o = 10.0\,\text{cm}$ from the lens. Find the image distance and magnification.

$$\frac{1}{d_i} = \frac{1}{15.0} - \frac{1}{10.0}$$

$$\frac{1}{d_i} = \frac{2-3}{30.0} = -\frac{1}{30.0}$$

$$\boxed{d_i = -30.0\,\text{cm}}$$

$$\boxed{m = -\frac{-30.0}{10.0} = +3.00}$$

$d_i < 0$: **virtual** image on the same side as the object.

$m > 0$: **upright**.

$|m| = 3$: **enlarged** threefold.

This is exactly how a magnifying glass works!

- Why can't a real image form when $d_o < f$?

Check Your Understanding

A converging lens with $f = 15\,\text{cm}$ has an object at $d_o = 10\,\text{cm}$. Is the image real or virtual? Upright or inverted? Larger or smaller than the object?

Come to lecture to see the answer.

Check Your Understanding

A lens produces an image with magnification $m = -3$. What does this tell you about the image?

Come to lecture to see the answer.

Part VII

Image Formation by Mirrors

Outline of Part VII: Image Formation by Mirrors

Outline of Part VII: Image Formation by Mirrors

Types of Mirrors

The Mirror Equation

Curved Mirrors

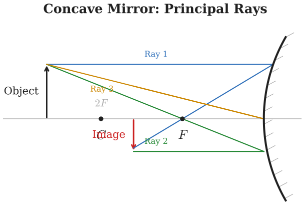
Three Mirror Types

- **Plane:** flat, $f = \infty$, virtual upright image
- **Concave:** inward curve, converging, $f > 0$
- **Convex:** outward curve, diverging, $f < 0$

Focal Length

$$\boxed{f = \frac{R}{2}}$$

R = radius of curvature. Real images form



Outline of Part VII: Image Formation by Mirrors

Types of Mirrors

The Mirror Equation

The Mirror Equation

Mirror Equation — (cp2e_ch25_s7)

Identical in form to the thin lens equation:

$$\boxed{\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f} = \frac{2}{R}}$$

Magnification: $m = h_i/h_o = -d_i/d_o$

Sign Conventions for Mirrors

- $d_i > 0$: real image, *in front* of mirror
- $d_i < 0$: virtual image, *behind* mirror
- $f > 0$: concave (converging)
- $f < 0$: convex (diverging)

Lens vs. Mirror

	Lens	**Mirror**
Mechanism	Refract.	Reflect.
Converging	$f > 0$	$f > 0$
Real image	Far side	Front face
Diverging	$f < 0$	$f < 0$

Concave Mirror Image Summary

Concave Mirror

Object	**Image**
Beyond $2f$	Real, inv., reduced
At $2f$	Real, inv., same size
Btwn f–$2f$	Real, inv., enlarged
At f	No image
Inside f	Virtual, upright, enl.

Applications

Concave: makeup mirrors, headlights

Convex: security, rear-view mirrors

Convex Mirror: Always

- Virtual
- Upright
- Reduced

Same pattern as a diverging lens.

Example: Concave Mirror — (cp2e_ch25_ex6)

Image in a Concave Mirror

A concave mirror has radius of curvature $R = 60.0\,\text{cm}$. An object is placed $d_o = 90.0\,\text{cm}$ in front. Find the image distance and magnification.

$$f = \frac{R}{2} = 30.0\,\text{cm}$$

$$\frac{1}{d_i} = \frac{1}{30.0} - \frac{1}{90.0}$$

$$\frac{1}{d_i} = \frac{3-1}{90.0} = \frac{2}{90.0}$$

$$\boxed{d_i = 45.0\,\text{cm}}$$

$$\boxed{m = -\frac{45.0}{90.0} = -0.500}$$

$d_i > 0$: **real** image in front of mirror.

$m < 0$: **inverted**.

$|m| = 0.5$: **reduced** to half size.

Object is beyond $2f$ ($90\,\text{cm} > 60\,\text{cm}$) — consistent with real, inverted, reduced. ✓

- If a bulb is placed at $f = 30\,\text{cm}$, where does the image form?

Example: Convex Security Mirror — (cp2e_ch25_ex7)

Convex Mirror

A convex security mirror has focal length $f = -0.500\,\mathrm{m}$. An object is $d_o = 3.00\,\mathrm{m}$ from the mirror. Find the image distance and magnification.

$$\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o}$$

$$\frac{1}{d_i} = \frac{1}{-0.500} - \frac{1}{3.00}$$

$$\frac{1}{d_i} = -2.00 - 0.333 = -2.333$$

$$\boxed{d_i \approx -0.429\,\mathrm{m}}$$

$$\boxed{m = -\frac{-0.429}{3.00} \approx +0.143}$$

$d_i < 0$: **virtual** image behind mirror.

$m > 0$: **upright**.

$|m| < 1$: **reduced**.

Wide field of view — useful for security and car mirrors.

- Why do convex mirrors always give upright, reduced images for real objects?

Check Your Understanding

The back of a spoon acts like a convex mirror. What type of image do you expect to see when you look at your reflection?

Come to lecture to see the answer.

Check Your Understanding

A makeup (concave) mirror with $f = 30\,\text{cm}$ is used with your face at $d_o = 20\,\text{cm}$. What kind of image forms, and why is this useful?

Come to lecture to see the answer.

Check Your Understanding

Why is the bulb in an automobile headlight placed at the focal point of a concave mirror reflector?

Come to lecture to see the answer.